AJKD
Atlas of Renal Pathology

Focal Segmental Glomerulosclerosis

Agnes Fogo, MD

The AJKD Atlas of Renal Pathology presents a compilation of figures on a specific pathologic entity. You may download the figures to create your own personal, non-commercial library of images or to create slides for teaching purposes.

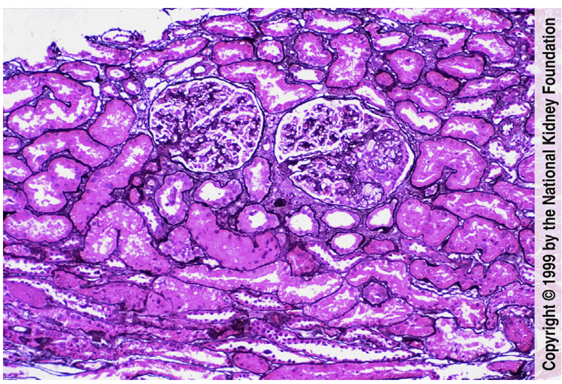

Fig 1. One of the two glomeruli shown contains two small peripheral foci of segmental sclerosis with intracapillary foam cells and prominence of overlying visceral epithelial cells. This is representative of an early segmental sclerosing lesion in focal segmental glomerulosclerosis (FSGS). There is also surrounding mild interstitial fibrosis, and the glomerulus that is sclerosed is mildly enlarged. Of note, significant glomerular-size increase, even without segmental sclerosis, is an indicator of possible unsampled or evolving FSGS. (Jones' silver stain, ×100).

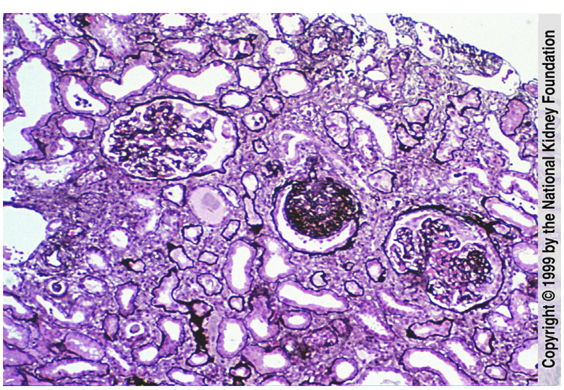

Fig 2. More advanced lesions of FSGS are present in this biopsy, with a central glomerulus showing global sclerosis, the glomerulus on the left showing well-defined peripheral segmental sclerosis, and the glomerulus on the right showing no lesions in this section. There is moderate interstitial fibrosis and tubular atrophy. The presence of globally sclerotic glomeruli does not aid in the specific diagnosis of FSGS, because global sclerosis may occur nonspecifically related to other injury processes or aging. (Jones' silver stain, ×100).

From the Department of Pathology, Vanderbilt University Medical Center, Nashville, TN.
Some cases kindly shared by Dr Robert G. Horn, Private Laboratory for Renal Biopsy Pathology, Nashville, TN; Medical Photographer: Brent Weedman; with assistance from Kim Solez, MD, of the National Kidney Foundation's cyberNephrology™ Team.
Address author queries to Agnes Fogo, MD, Department of Pathology, Vanderbilt University Medical Center, MCN C-3310, Nashville, TN 37232. E-mail: Agnes.Fogo@vanderbilt.edu
Am J Kidney Dis. 33(4):E1-E2.1.

0272-6386/$36.00
http://dx.doi.org/10.1053/S0272-6386(13)90015-9

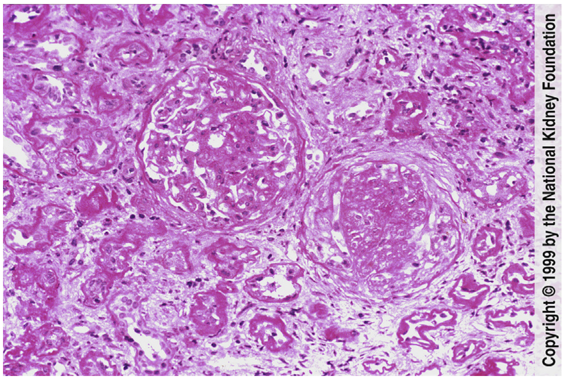

Fig 3. Advanced segmental sclerotic lesions are present in these two glomeruli, with increased mesangial matrix and obliteration of capillaries and with more extensive lesion in the glomerulus on the right. The lesions are typical of idiopathic FSGS. The interstitium show moderate to severe fibrosis and tubular atrophy. (Periodic acid-Schiff stain, ×200).

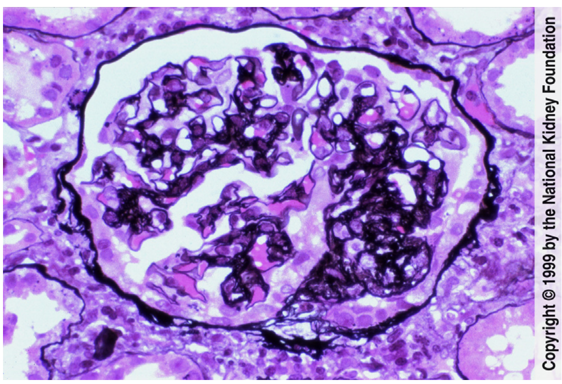

Fig 4. In this biopsy, the segmental sclerotic lesion on the right with FSGS is composed of obliteration of capillary lumens and increased mesangial matrix. The remaining portion of the glomerular tuft appears unremarkable, without basement membrane changes or any apparent deposits. The visceral epithelial cells overlying this early sclerotic lesion appear activated. (Jones' silver stain, ×400).

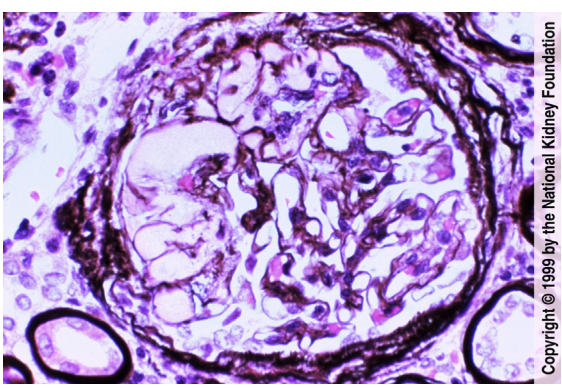

Fig 5. This glomerulus shows segmental sclerosis with hyalinosis, typical of FSGS. Hyalin is defined as smooth, glassy-appearing material, resulting from the insudation of plasma proteins. It frequently occurs in sclerotic lesions, regardless of primary origin, and is not of primary diagnostic significance. It is frequently observed in cases of FSGS, whether primary or secondary. (Jones' silver stain, ×400).

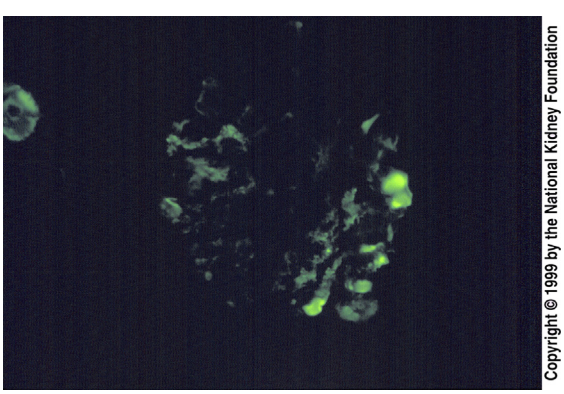

Fig 6. Immunofluorescence studies in cases of FSGS may show immunoglobulin (Ig) M in mesangial areas or in areas of hyalinosis (smooth, globular staining on right). These may be associated with small areas of electron density by electron microscopy, although typical immune-complex densities are not found. The IgM staining is thought to represent trapping in areas of expanded mesangium and sclerosis. (Immunofluorescence with anti-IgM, ×200).

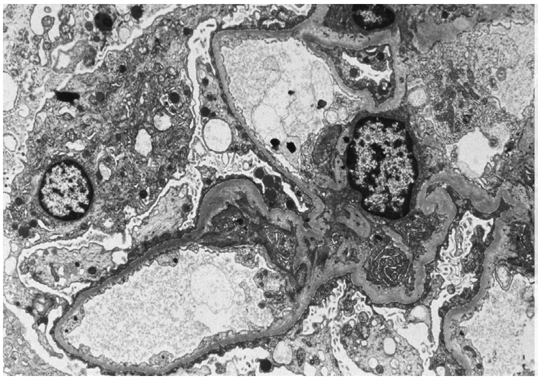

Fig 7. The mesangial matrix is mildly increased without deposits in this case of FSGS. Endothelial cells are unremarkable, and visceral epithelial cells show extensive blunting and effacement of foot processes with early microvillous transformation. Occasional vacuolization and blebbing is also present in the visceral epithelial cells. No immune complexes are present. (Transmission electron microscopy, ×3,000).